

Workforce Data Requirements Document

Project Title: AI-Powered Workforce Analytics & Talent Intelligence Dashboard

1. Document Metadata

Attribute	Details
Document Name	Workforce Data Requirements Document
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Milestone	Milestone 1 – Task 1 (Foundation for Data Repository)
Status	Approved / Baseline
Target Audience	HR Executives, System Architects, ETL Engineers, Data Governance Team

2. Executive Summary

This document establishes the comprehensive workforce data requirements for the **AI-Powered Workforce Analytics & Talent Intelligence Dashboard**. The primary goal of this initiative is to create a secure, centralized, and scalable workforce data repository that powers real-time analytics, predictive modeling (attrition, performance, recruitment), and conversational intelligence using Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and agentic AI workflows. This specification captures key business needs gathered through stakeholder interviews with CHROs, Talent Acquisition Leads, HR Business Partners, and Enterprise Risk/IT teams.

3. Business Objectives & Strategic Value

- **Data-Driven Strategic Workforce Planning:** Consolidate disparate data streams across SAP HR, Workday, ATS, and LMS to eliminate data silos.
- **Proactive Talent Retention:** Identify early attrition risks and turnover drivers through real-time and historical workforce metrics.
- **Conversational Querying & Agentic AI:** Enable executives to query complex workforce

analytics in natural language via LLMs and RAG integration.

- **Diversity, Equity & Inclusion (DEI) Visibility:** Provide holistic tracking of diversity metrics across pay equity, promotion velocity, and leadership pipelines.

4. Stakeholder Identification & Analysis

Stakeholder Role	Key Business Goals & Needs	Required Outputs & Intelligence
Chief Human Resources Officer (CHRO) & C-Suite	Macro-level organizational health, strategic headcount planning, retention trends, and executive summaries.	High-level KPI dashboards, attrition forecasting models, natural language executive briefing summaries.
HR Business Partners (HRBPs)	Departmental performance tracking, employee engagement analysis, manager-level attrition analysis.	Filterable department dashboards, manager scorecard reports, sentiment trend analysis.
Talent Acquisition Leads	Recruitment funnel efficiency, time-to-fill, cost-per-hire, and sourcing quality.	Applicant tracking metrics, candidate drop-off funnel, recruitment ROI reports.
Learning & Development (L&D) Managers	Skill gaps identification, training completion tracking, and internal mobility planning.	Skill inventory matrices, training impact vs. performance ratings, succession planning feeds.
Data Security & Compliance Officers	Ensure strict compliance with PII regulations, GDPR, role-based access control, and audit trail generation.	Anonymized datasets, data access logs, data protection compliance validation reports.

5. Functional Workforce Data Requirements

To support downstream analytics and AI workflows, the repository must ingest, structure, and store data across five core domains:

5.1 Employee Demographics & Core HR Data

- **Employee Identification:** Employee ID, Full Name, Gender, Age, Marital Status.
- **Organizational Context:** Department, Job Role, Job Level, Manager ID, Business Unit, Location/Geography.
- **Tenure & Status:** Hire Date, Original Hire Date, Employment Status (Active/Terminated), Full-Time/Part-Time status.

5.2 Compensation & Payroll Analytics

- **Pay Rates:** Daily Rate, Hourly Rate, Monthly Rate, Monthly Income, Annual Salary.
- **Equity & Growth:** Stock Option Level, Percent Salary Hike, Bonus History, Overtime Status (Yes/No).
- **Standard Hours:** Standard Hours baseline per region/role.

5.3 Performance, Engagement & Satisfaction

- **Performance Ratings:** Manager Performance Rating, Self-Rating, Review Period.
- **Satisfaction Scores:** Job Satisfaction, Environment Satisfaction, Relationship Satisfaction, Work-Life Balance score (1-4 scale).
- **Involvement & Engagement:** Job Involvement score, pulse survey ratings, tenure in current role, years with current manager.

5.4 Attrition & Organizational Dynamics

- **Attrition Status:** Attrition Flag (Yes/No), Separation Date, Exit Reason, Voluntary vs. Involuntary termination.
- **Work History:** Total Working Years, Number of Companies Worked, Years at Company, Years Since Last Promotion.
- **Commute & Travel:** Distance From Home (miles/km), Business Travel Frequency (Non-Travel, Travel_Rarely, Travel_Frequently).

5.5 Talent Development & Training

- **Learning Metrics:** Training Times Last Year, Education Level (1-5), Education Field.
- **Skill Inventories:** Primary Skillsets, Competency Certifications, Future Readiness Scores.

6. Target Data Sources & Ingestion Scope

Source System	Target Entities	Frequency of Ingestion	Integration Mode
SAP HR / SuccessFactors	Core Employee Profiles, Payroll Data, Global Compensation,	Daily Batch / Incremental Delta	REST API / OData Services

Source System	Target Entities	Frequency of Ingestion	Integration Mode
	Organizational Hierarchy		
Workday HCM	Job Architecture, Time Off, OverTime Data, Absence Management	Daily Batch	Workday RaaS / SOAP API
Applicant Tracking System (ATS)	Candidate Pipelines, Requisition Status, Offer Acceptance Rates, Sourcing Channels	Real-Time / Hourly API Sync	Webhook / REST API
Learning Management System (LMS)	Course Completions, Training Hours Last Year, Skill Assessment Scores	Weekly Batch	CSV / Automated SFTP Ingestion

7. Security, Privacy & Data Governance Requirements

- **Personally Identifiable Information (PII) Protection:** Names, Social Security / National ID numbers, and direct contacts must be masked or anonymized in general analytics layers.
- **Role-Based Access Control (RBAC):** Managers should only view aggregated/individual data for their direct and indirect reports. Executives receive organization-wide aggregated metrics.
- **AI Safety & Vector Store Security:** RAG vector embeddings must strictly inherit underlying data access permissions so that AI agents do not leak confidential executive pay or termination reasons.
- **Regulatory Compliance:** Adherence to GDPR, CCPA, and regional labor data storage mandates.